

Fourier interpolation of the q -space TDSCHA Lanczos without high-order force-constant tensors

TDSCHA development notes — `qspace_interpolation` branch

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Abstract

This note explains how to run the q -space TDSCHA Lanczos response on a fine uniform q mesh while the stochastic ensemble remains in a smaller coarse supercell. The physical motivation is simple: the response vector contains two-phonon blocks, and those blocks should sample a dense two-phonon continuum even when the expensive force calculations were done only in a coarse cell.

The central constraint is that we must not build high-order force-constant tensors. Instead, we interpolate the *per-configuration Bloch fields* that already enter the stochastic TDSCHA estimator. Windowing those fields induces a real-space image-assignment kernel on the averaged three- and four-field correlations. The report derives this kernel explicitly, shows why the plain off-grid Bloch sum is biased between commensurate points, and designs windows that reproduce the same minimal-image logic as centered tensor interpolation while keeping the runtime linear in the number of fine-mesh points.

Three invariants organize the construction. First, at commensurate wavevectors the interpolated operator must reduce exactly to the original coarse q -space Lanczos operator. Second, the interpolation kernel must respect the relevant permutation symmetries of the stochastic estimator. Third, acoustic sum rules must hold at the vertex level. We show how these conditions become linear constraints on the window kernel. For the third order this gives a tensor-free atom-resolved centering that reproduces the SnTe $2^3 \rightarrow 4^3 \Gamma$ TO benchmark. For the fourth order the report documents why two simpler D4 centering attempts fail on SnTe and proposes a new four-leg, ASR-projected, factorized D4 kernel. The proposed D4 factorization is a low-rank representation of the *geometry kernel*, not of $\Phi^{(4)}$; the report proves its commensurate identity and its equivalence to the stochastic q -space D4 contraction.

Contents

1	Notation and problem statement	2
2	Background: the q-space TDSCHA Lanczos	3
3	Off-grid Bloch fields and the effective interpolation kernel	4
3.1	Why the naive off-grid transform is not an interpolation	4
3.2	The effective kernel of a windowed estimator	4
3.3	Equivalence with centered-tensor interpolation	5
4	Stochastic centering: designed multitaper windows	5
4.1	Exactly solvable case: pair kernels	5
4.2	Triple kernels: low-rank ALS design	5
5	Vertex renormalization $N_c \rightarrow N_f$	6
6	Field pre-filtering: interpolating Φ-ranged objects	6

7 Acoustic sum rules: a one-shot centering + ASR + symmetrization construction	7
7.1 What the ASR means at the kernel level	7
7.2 A no-go theorem for one-period windows	8
7.3 The one-shot construction: doubled support, uniform class sums	8
7.4 Where the leak actually bites, and validation	9
7.5 Response-level validation: leak versus resolution	10
8 Atom-resolved pinned centering for $L = 2$ real-material cells	11
9 Implementation and cost	13
10 Benchmarks	14
11 Example application: spectral functions at interpolated q and the standard d_3 bubble	15
12 Fourth order: spectral functions where D_4 is fundamental	16
12.1 Atomic centering of the four-phonon vertex	18
13 Limitations and outlook	22
A Derivation of the effective interpolation kernel	23
B Gaussian integration by parts and the pre-filter identity	23
C Reproducibility	24

1 Notation and problem statement

This section fixes the notation used throughout the report and states the problem in operational terms.

symbol	meaning
N_c	number of cells, hence number of commensurate q points, in the coarse ensemble supercell
N_f	number of points in the fine interpolation mesh
n_{at}	number of atoms in the primitive cell
$n_b = 3n_{\text{at}}$	number of phonon bands at each q point
I	stochastic configuration index
$\mathbf{R}$	coarse-supercell lattice vector
$a, b, c, \dots$	atom and Cartesian indices, often collapsed into one index
$\mathbf{q}_p$	perturbing phonon wavevector of the response calculation
$\mathbf{q}_1, \mathbf{q}_2$	two-phonon wavevectors satisfying $\mathbf{q}_1 + \mathbf{q}_2 = \mathbf{q}_p + \mathbf{G}$
$x_\nu(\mathbf{q}; I)$	mass-weighted displacement Bloch field in phonon mode ν
$y_\nu(\mathbf{q}; I)$	mass-weighted force-residual Bloch field in phonon mode ν
f_ψ	SSCHA displacement covariance factor $(1 + 2n)/2\omega$
f_Υ	inverse covariance factor $2\omega/(1 + 2n)$
$\Phi^{(n)}$	n -th order force-constant tensor, used only for comparison, not stored by the interpolation algorithm

The input data are the SSCHA ensemble configurations and forces in the coarse supercell, plus the converged SSCHA dynamical matrix. The desired output is the same q -space TDSCHA

Lanczos operator that would be obtained from a direct fine-supercell ensemble, but evaluated from the coarse data. The important design restrictions are:

1. never construct or store $\Phi^{(3)}$ or $\Phi^{(4)}$;
2. keep setup and each Lanczos application linear in N_f ;
3. reproduce the original q-space operator exactly when the fine point is commensurate with the coarse mesh;
4. preserve Hermiticity, permutation symmetries, and acoustic sum rules.

The report is organized as follows. Sections 2–6 derive the stochastic interpolation kernel. Sections 7–8 add ASR and atom-resolved centering. Section 9 describes the implementation and cost. Sections 10–12 give the tests, including the current state of the fourth-order problem. The appendices repeat the two core derivations in a compact form.

2 Background: the q-space TDSCHA Lanczos

The q-space TDSCHA Lanczos [1, 2] computes a response function by repeatedly applying the linearized TDSCHA superoperator $\mathcal{L}$ to a vector with one one-phonon component and many two-phonon components. For a perturbation at $\mathbf{q}_p$ the vector has the schematic form

$$\psi = [R_\nu(\mathbf{q}_p) | a'_{\nu_1\nu_2}(\mathbf{q}_1, \mathbf{q}_2) | b'_{\nu_1\nu_2}(\mathbf{q}_1, \mathbf{q}_2)], \quad \mathbf{q}_1 + \mathbf{q}_2 = \mathbf{q}_p + \mathbf{G}, \quad (1)$$

with one $n_b \times n_b$ block for each pair $(\mathbf{q}_1, \mathbf{q}_2)$ in the mesh. The harmonic part is simple once the SSCHA frequencies $\omega_\nu(\mathbf{q})$ and polarization vectors $e_\nu(\mathbf{q})$ are known. The only difficult part is the anharmonic action.

For each ensemble configuration I with statistical weight ρ_I , the code forms two Bloch fields:

$$\begin{aligned} x_\nu(\mathbf{q}; I) &= \sum_a e_\nu^{a*}(\mathbf{q}) \sqrt{m_a} \tilde{u}_a(\mathbf{q}; I), & \tilde{u}_a(\mathbf{q}; I) &= \frac{1}{\sqrt{N_c}} \sum_{\mathbf{R}} e^{-i\mathbf{q}\cdot\mathbf{R}} u_a(\mathbf{R}; I), \\ y_\nu(\mathbf{q}; I) &= \sum_a e_\nu^{a*}(\mathbf{q}) \frac{1}{\sqrt{m_a}} \delta \tilde{f}_a(\mathbf{q}; I), & \delta f &= f - f_{\text{SSCHA}} - \langle f \rangle, \end{aligned} \quad (2)$$

where $\mathbf{R}$ runs over the N_c coarse-cell origins and the phase convention matches `vector.r2q`. The field x carries displacements; the field y carries force residuals. The Julia fused kernel contracts these fields into the TDSCHA action. Every anharmonic term is an ensemble average of either three fields (the D_3 part) or four fields (the D_4 part). For example, with $f_\Upsilon(\mathbf{q}, \nu) = 2\omega/(1+2n)$,

$$\left\langle \overline{(f_\Upsilon x)}(\mathbf{q}_p) x(\mathbf{q}_1) y(\mathbf{q}_2) \right\rangle, \quad \langle \bar{y}(\mathbf{q}_p) x(\mathbf{q}_1) x(\mathbf{q}_2) \rangle, \quad \dots \quad (3)$$

are representative D_3 averages, and the D_4 averages contain one more displacement field.

These averages are not heuristic. They are exact Gaussian integration-by-parts estimators of the anharmonic vertices. For instance, for the Gaussian SSCHA density with covariance Υ^{-1} ,

$$\langle u_a u_b \delta f_c \rangle = \Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \langle \partial^3 V / \partial R_{a'} \partial R_{b'} \partial R_c \rangle, \quad (4)$$

up to the sign convention of the force. Disconnected Wick pieces vanish at SSCHA stationarity because $\langle u \rangle = 0$, $\langle \delta f \rangle = 0$, and $\langle u \otimes \delta f \rangle \propto \Upsilon^{-1}(\langle \partial^2 V \rangle - \Phi) = 0$. The same argument with one more displacement field gives the D_4 estimator. This is the key fact used throughout the report: the code estimates contractions of $\Phi^{(3)}$ and $\Phi^{(4)}$, but never needs to materialize those tensors.

3 Off-grid Bloch fields and the effective interpolation kernel

3.1 Why the naive off-grid transform is not an interpolation

The ensemble configuration is periodic with the coarse supercell. It therefore contains exact Fourier information only at the N_c commensurate wavevectors of that supercell. If one periodically repeats the configuration to infinity and Fourier transforms it at a genuinely incommensurate $\tilde{\mathbf{q}}$, the result is zero: the spectrum is a comb of delta functions at the commensurate points.

The code can nevertheless evaluate the finite sum in Eq. (2) at any $\tilde{\mathbf{q}}$. This is not wrong, but it has a precise meaning: it is a Fourier transform of one supercell multiplied by a rectangular window. The output is controlled by how this window assigns the periodic images of each real-space correlation to the phase factor $e^{-i\tilde{\mathbf{q}}\cdot\mathbf{R}}$. This is the same issue that tensor interpolation solves by centering: before Fourier transforming $\Phi^{(n)}$, one chooses the periodic image of every tensor element that has the shortest real-space geometry, and then zero-pads the rest.

The goal of stochastic centering is to reproduce that image assignment on the estimator itself. The fields are windowed before entering the stochastic averages; after ensemble averaging, those windows become an explicit real-space interpolation kernel.

3.2 The effective kernel of a windowed estimator

Consider first one D_3 product. The three slots are labelled (z, w, v) : the z slot carries the perturbation wavevector, and the w, v slots carry the two-phonon pair. Give each slot its own real window over the coarse-cell label $\mathbf{n}$:

$$\tilde{x}_W(\tilde{\mathbf{q}}; I) \propto \sum_{\mathbf{n}} W(\mathbf{n}) e^{-i\tilde{\mathbf{q}}\cdot\mathbf{R}_{\mathbf{n}}} u(\mathbf{R}_{\mathbf{n}}; I). \quad (5)$$

The window is zero outside its chosen support; the plain estimator is the special case $W \equiv 1$ on the whole coarse period. For a fine mesh that contains $\mathbf{q}_p$, the pair map enforces $\tilde{\mathbf{q}}_1 + \tilde{\mathbf{q}}_2 = \tilde{\mathbf{q}}_p + \mathbf{G}$ exactly. Translation invariance of the ensemble then gives

$$\langle \tilde{z}(\tilde{\mathbf{q}}_p) \tilde{x}_w(\tilde{\mathbf{q}}_1) \tilde{y}_v(\tilde{\mathbf{q}}_2) \rangle = \frac{1}{N_c^{3/2}} \sum_{\delta_1 \delta_2} S_{z,w,v}(\delta_1, \delta_2) c_3(\delta_1, \delta_2) e^{-i\tilde{\mathbf{q}}_1 \cdot \delta_1 - i\tilde{\mathbf{q}}_2 \cdot \delta_2}, \quad (6)$$

where c_3 is the physical three-point correlation at *actual* (unwrapped) lattice differences, the sum runs over all periodic images, and the **effective interpolation kernel** is the (per-dimension separable) window triple correlation

$$S_{z,w,v}(\delta_1, \delta_2) = \sum_s z(s) w(s + \delta_1) v(s + \delta_2). \quad (7)$$

Here δ_1 and δ_2 are actual, unwrapped lattice differences from the z slot to the w and v slots. The correlation $c_3(\delta_1, \delta_2)$ is the physical real-space three-point correlation. The function $S_{z,w,v}$ is the image-assignment kernel created by the windows. This equation is the main mathematical reduction of the report: interpolation quality is kernel quality.

Two consequences are immediate.

1. *Partition of unity.* For the plain window, $\sum_{\text{images}} S(\delta + L\mathbf{n}) = L$ for every difference class per dimension of length L . At commensurate $\tilde{\mathbf{q}}$, all images of the same folded class carry the same phase, so only this image sum matters. The estimator then reduces exactly to the coarse one.
2. *Tent bias off-grid.* The plain kernel assigns weight L – spread to each image: for $L = 4$, a nearest-neighbour correlation component leaks 25% of its weight onto an image three cells away, which enters with the wrong phase at incommensurate $\tilde{\mathbf{q}}$. The interpolation passes through the exact on-grid values but sags in between.

3.3 Equivalence with centered-tensor interpolation

Equation (6) states that the ensemble expectation of the off-grid estimator *is* the trigonometric (Fourier) interpolation of the real-space correlation function c_3 , with image weights S . Tensor interpolation performs the same Fourier sum after centering the tensor entries. Therefore:

If the stochastic kernel S equals the minimal-image indicator used by tensor centering, including tie splitting, then the expectation value of the windowed estimator is identical to the centered-tensor interpolation.

The difference is the order of operations. Tensor interpolation first forms a high-order real-space object and then contracts it. The stochastic method applies windows to the configurations and contracts the result configuration by configuration. The same centered object is obtained in the infinite-sampling limit, but the tensor is never stored. With the pre-filtering of Sec. 6, the interpolated correlation has the range of $\Phi^{(3)}$ itself, so this comparison is with the same short-ranged object used by `cellconstructor.Spectral`.

4 Stochastic centering: designed multitaper windows

4.1 Exactly solvable case: pair kernels

The previous section reduced the problem to kernel design. Before fitting the three-field kernel, it is useful to look at a two-field example where the answer is analytic. For a one-dimensional coarse period of length L , the minimal-image pair kernel is obtained exactly from two rectangular windows. With $r_m(\delta) = \max(0, m - |\delta|)$ the autocorrelation of $\text{rect}(m)$, for even L

$$\frac{1}{2} [r_{L/2+1}(\delta) - r_{L/2-1}(\delta)] = \begin{cases} 1 & |\delta| \leq L/2 - 1 \\ 1/2 & |\delta| = L/2 \quad (\text{Wigner-Seitz tie}) \\ 0 & \text{beyond,} \end{cases} \quad (8)$$

including the correct tie splitting, and satisfying the partition of unity. This simple identity proves the principle: sharp minimal-image centering can be represented by separable windows applied to configurations, not only by explicit tensors.

4.2 Triple kernels: low-rank ALS design

For D_3 the target depends on two differences, because the three legs form a cluster. In one dimension define

$$\text{spread}(0, \delta_1, \delta_2) = \max(0, \delta_1, \delta_2) - \min(0, \delta_1, \delta_2).$$

The target $M(\delta_1, \delta_2)$ assigns weight to the image(s) with minimal spread and splits exact ties. This is the separable one-dimensional version of `ForceTensor`'s d_3 centering.

Using the same window in all three slots cannot improve on the tent kernel. The useful freedom is to allow the three slots to carry different windows. For each dimension we therefore fit K passes (z_r, w_r, v_r) by alternating least squares:

$$\sum_{r=1}^K \frac{1}{2} [S_{z_r, w_r, v_r} + S_{z_r, v_r, w_r}^T] \approx L M, \quad \text{subject to} \quad \sum_{\text{images}} \sum_r S_r = L \quad (\text{hard partition constraint}). \quad (9)$$

The transpose average enforces the $w \leftrightarrow v$ symmetry of the two-phonon block. The hard partition constraint is what protects the exact commensurate limit. For the sizes used in the tests the fitted kernels are essentially exact ($L = 3, K = 3$: residual 2×10^{-16} ; $L = 4, K = 2$: 8×10^{-12}), as shown in Fig. 1.

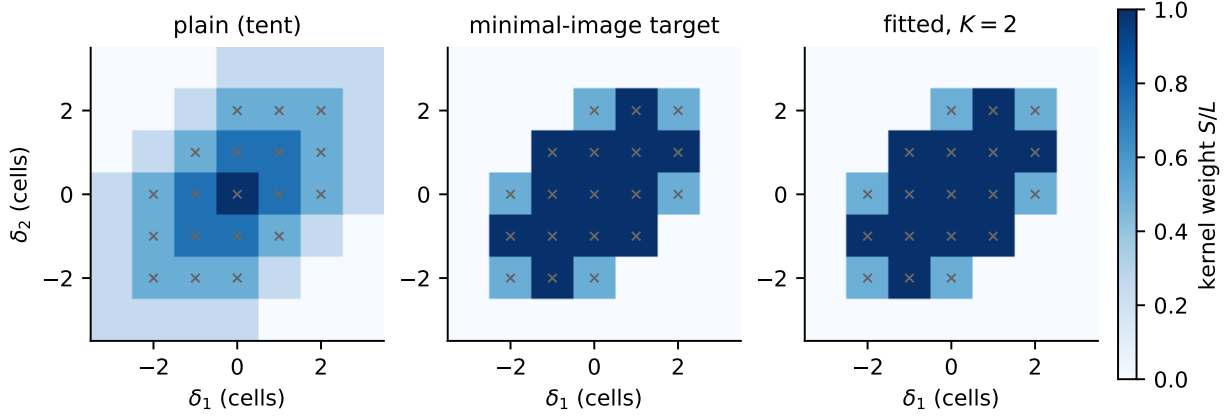


Figure 1: Effective interpolation kernels on the difference lattice (δ_1, δ_2) for a coarse dimension $L = 4$ (single hue: darker = larger weight; grey crosses mark the exact-image positions of the minimal-image target with their tie weights). Left: plain (tent) kernel of the naive off-grid transform. Middle: minimal-image target = zero-padded centering. Right: fitted $K = 2$ multitaper kernel — indistinguishable from the target (8×10^{-12} RMS).

In three dimensions the windows are products of one-dimensional factors, so K one-dimensional passes become K^3 three-dimensional passes. Each pass is just one call to the same Julia stochastic kernel. Optional origin averaging reduces variance; it must be implemented by rolling the configuration data while leaving the window support fixed. Shifting the window itself would wrap its support and would change the intended zero-padding.

5 Vertex renormalization $N_c \rightarrow N_f$

Changing the two-phonon mesh changes the normalization of mode-space vertices. A vertex written on an N -cell mesh scales as

$$d_3(\mathbf{q}_1\nu_1, \mathbf{q}_2\nu_2, \mathbf{q}_3\nu_3) = \frac{1}{\sqrt{N}} \delta_{\mathbf{q}_1+\mathbf{q}_2+\mathbf{q}_3, \mathbf{G}} \bar{\Phi}_3(\dots), \quad d_4 = \frac{1}{N} \delta_{\Sigma\mathbf{q}, \mathbf{G}} \bar{\Phi}_4, \quad (10)$$

where $\bar{\Phi}_n$ is intensive. The stochastic fields come from the coarse cell, so their raw averages carry the $N = N_c$ normalization. The interpolated Lanczos operator, however, is meant to act as the operator of an N_f -point fine mesh. Therefore the vertex factors must be converted:

$$s_3 = \sqrt{N_c/N_f}, \quad s_4 = N_c/N_f.$$

These are the `scale3` and `scale4` arguments of the Julia kernel.

A useful check is the one-loop bubble. The bubble has N_f pair terms, each containing $|d_3|^2$. With the scale factor, $|d_3|^2$ carries $s_3^2/N_c = 1/N_f$, so the total bubble is intensive and matches the direct fine-mesh normalization. The uniform per-order rescaling is exact up to disconnected Wick pieces $\propto \langle u \otimes \delta f \rangle$, which vanish at SSCHA stationarity; the ensemble should therefore be reweighted to the converged SSCHA dynamical matrix before interpolating. Empirically, *disabling* the rescaling makes the anharmonic renormalization overshoot by a factor $\simeq N_f/N_c$ (pinned by a regression test), confirming both its presence and magnitude.

6 Field pre-filtering: interpolating Φ -ranged objects

The windows should interpolate a short-ranged object. The raw stochastic correlations are not short-ranged enough, because each displacement leg carries the SSCHA covariance factor $f_\psi =$

$(1 + 2n)/2\omega$. For example,

$$\langle x(\mathbf{q}_1) x(\mathbf{q}_2) y(\mathbf{q}_3) \rangle = f_\psi(\mathbf{q}_1) f_\psi(\mathbf{q}_2) [\Phi^{(3)} \text{ contraction}]. \quad (11)$$

In real space this correlation decays with the propagator range, including the acoustic $1/\omega$ tail, rather than with the range of $\Phi^{(3)}$. In the chain benchmark this caused a systematic underestimate of the anharmonic renormalization by about a factor two.

The fix is to move the covariance inverse $f_\Upsilon = 1/f_\psi$ from the Julia kernel into the coarse-grid fields before interpolation:

1. apply f_Υ exactly on the coarse grid, mode by mode, for each configuration;
2. transform the resulting filtered real-space configuration to the fine wavevectors;
3. set the kernel tables $f_\Upsilon, f_\psi \rightarrow 1$ (**prefiltered** flag);
4. fold the exact fine-side $f_\psi(\tilde{\mathbf{q}}_1) \otimes f_\psi(\tilde{\mathbf{q}}_2)$ into the $\alpha^{(1)}$ blocks (the α -contracted legs also use the filtered fields).

At commensurate points this is an exact rewriting of the old estimator, so backward compatibility is unchanged. Off-grid, the object being windowed now has the range of the force constants themselves. Frequencies, polarizations, χ , f_Υ and f_ψ are always evaluated from the interpolated SSCHA dynamical matrix on the fine mesh; no band-structure quantity is inferred stochastically.

7 Acoustic sum rules: a one-shot centering + ASR + symmetrization construction

Centering and the acoustic sum rule pull in different directions. A minimal-image kernel chooses compact images; an ASR-correct kernel must also vanish when any vertex leg is a uniform translation. In tensor codes this is why `ForceTensor.Center()` is followed by `Apply_ASR()`.

For the stochastic estimator the issue is even more concrete. Before windowing, the total force and the center-of-mass displacement constraints hold configuration by configuration. A nonuniform window cuts out part of the supercell, so those exact cancellations are no longer automatic. The problem is therefore to design a window kernel that:

1. has the desired centered image assignment off the coarse grid;
2. has exact partition sums, hence exact commensurate identity;
3. has zero coupling to an acoustic translation on any leg;
4. keeps the $w \leftrightarrow v$ permutation symmetry of the pair block.

The useful observation is that all these requirements are linear constraints on the kernel S .

7.1 What the ASR means at the kernel level

Let ϕ_{per} denote the coarse-periodic three-leg object being interpolated after pre-filtering. Translational invariance says that summing over any one physical leg gives zero. In difference coordinates and one lattice dimension of length L , this becomes the set of period sums

$$\sum_{d=0}^{L-1} \phi_{\text{per}}(d, \cdot) = 0, \quad \sum_{d=0}^{L-1} \phi_{\text{per}}(\cdot, d) = 0, \quad \sum_{d=0}^{L-1} \phi_{\text{per}}(d, d + \Delta) = 0 \quad \forall \Delta, \quad (12)$$

The third sum looks diagonal because the first leg is the reference leg of the difference coordinates. The interpolated object is the pointwise product $S \phi_{\text{per}}$ on the extended image grid. It satisfies the ASR for every possible ϕ_{per} obeying Eq. (12) if the image sums of S are class-independent:

$$\begin{aligned} (\text{ASR-}v) \quad & \sum_b S(\delta_1, d_2 + bL) \text{ independent of } d_2 \text{ for every fixed } \delta_1, \\ (\text{ASR-}w) \quad & \sum_a S(d_1 + aL, \delta_2) \text{ independent of } d_1 \text{ for every fixed } \delta_2, \\ (\text{ASR-}z) \quad & \sum_b S(d + bL, d + \Delta + bL) \text{ independent of } d \text{ for every fixed } \Delta. \end{aligned} \quad (13)$$

These are linear constraints on S . They are the kernel version of **Apply_ASR**. They also contain the partition condition: once the class sums are constant in each leg, only one scalar normalization is needed to recover the commensurate operator. Two checks follow.

- The plain (tent) kernel satisfies Eq. (13) exactly (its per-configuration sum rules are inherited untouched): the baseline estimator is ASR-clean everywhere.
- The *minimal-image kernel violates Eq. (13)* — the kernel-space restatement of “centering destroys the sum rule”, i.e. the very reason **Apply_ASR** exists. No admissible kernel that is exactly minimal-image can be ASR-exact at compact support.

7.2 A no-go theorem for one-period windows

If a window is supported on exactly one coarse period, its folded class sum is just the window value itself. Uniform class sums therefore imply the plain window. This proves a no-go statement: one-period nontrivial windows can improve centering, but they cannot also be ASR-exact at the kernel level.

A field-level projection is not a clean workaround. Subtracting a window-weighted acoustic zero mode modifies the effective window by a rank-one correction. That means the runtime kernel is no longer the kernel that was fitted. On the toy model this produced an $O(30\%)$ systematic bias. This is why the implementation applies such projections only for uniform or near-uniform windows.

7.3 The one-shot construction: doubled support, uniform class sums

The obstruction disappears if each window is allowed to live on two periods. The data are still periodic, but the Bloch phases are not wrapped: this is genuine zero-padding, just as in tensor centering. Parameterize a one-dimensional window by

$$w(d) = h(d), \quad w(d + L) = c_w - h(d), \quad d = 0 \dots L - 1, \quad (14)$$

with $h \in \mathbb{R}^L$ and c_w free. The folded class sum is $w(d) + w(d + L) = c_w$, independent of d . Thus uniform class sums are imposed by parameterization, not by a penalty, while the window can still be nontrivial.

On this manifold:

1. (ASR- v) and (ASR- w) hold identically; (ASR- z) is found to hold automatically for every fitted design (10^{-15} , no extra constraint needed).
2. Every per-pass kernel has *identical* class sums $L c_z c_w c_v$, so the full partition of unity (L^2 constraints) collapses to the single scalar equation $\sum_r c_z^r c_w^r c_v^r = 1$, enforced as a hard KKT row in each ALS slot fit: the commensurate limit is exact to machine precision. Indeed at commensurate $\tilde{q}$ the second-period Bloch phase is 1, each pass collapses to (class-sum product) $\times$ plain *configuration by configuration*, and the pass sum reproduces the plain estimator exactly (verified at 10^{-22}).

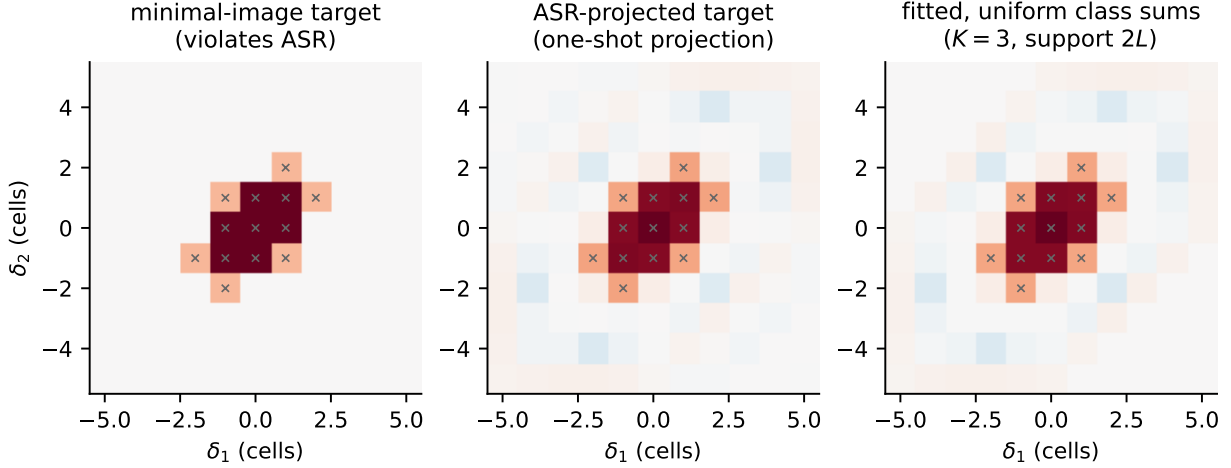


Figure 2: One-shot ASR-exact centering at $L = 3$ (diverging scale, red positive / blue negative; grey crosses = exact-image positions). Left: minimal-image target, which violates the kernel sum rules Eq. (13). Middle: its closed-form projection onto the ASR + partition subspace. Right: kernel realized by $K = 3$ doubled-support window passes with exactly uniform class sums — partition, all three sum rules and the $w \leftrightarrow v$ symmetry hold at 10^{-15} simultaneously.

3. At $\tilde{q} = 0$ the effective weight is the constant class sum, so the acoustic contraction vanishes *per configuration with zero statistical noise* — the coarse-estimator mechanism restored. No field projection is needed or applied.
4. The $w \leftrightarrow v$ symmetrization is part of the kernel definition (orientation averaging), so it commutes with everything above.

This is the reason the stochastic kernel can be fitted in one shot. The centering target, ASR, partition, and symmetry are simultaneous linear conditions on the small kernel. In tensor space, by contrast, centering, ASR, and symmetrization are usually applied as separate projections.

The target cannot be the raw minimal-image kernel, because that kernel violates Eq. (13). The correct target is the projection of the minimal-image kernel onto the ASR and partition subspace. The projection distance is the unavoidable price of imposing exact ASR at finite support, analogous to the correction made by `Apply_ASR`:

RMS distance $LM \rightarrow$ projection	$L = 3$	$L = 4$	$L = 6$
support $2L$	0.144	0.248	0.374
support $3L$	0.072	0.127	0.193
plain (tent) kernel, for scale	0.315	0.462	—

The fitted windows reproduce the projected target nearly exactly (residual 0.028 at $L = 3$, $K = 3$; 0.071 at $L = 4$). Increasing support improves centering fidelity while preserving exact ASR. An optional metric weight ρ^{spread} can concentrate fidelity at short range, analogous to choosing how `Apply_ASR` spreads its correction. At runtime a doubled-support window on periodic data becomes the per- q complex weight $\hat{w}_{\tilde{q}}(s) = w(s) + w(s+L) e^{-2\pi i \tilde{q} \cdot L \mathbf{a}}$ per dimension. No change to the Julia kernels is required.

7.4 Where the leak actually bites, and validation

Contracting each kernel with an exact ASR-satisfying test tensor gives the leak *deterministically* (Fig. 3, left): the translation contraction of the interpolated vertex on the acoustic leg $\tilde{q}_2 \rightarrow 0$ at fixed partner $\tilde{q}_1$. Two structural facts emerged that reshape where one should look for ASR artifacts:

- *The leak vanishes when both pair legs go to Γ together ($q_p = 0$):* the minimal-image violation carries phases that cancel at $\tilde{q}_1 \rightarrow 0$, and it re-vanishes at every commensurate $\tilde{q}_1$. The dangerous channel is an acoustic leg at fixed *finite* $\tilde{q}_1$ — physically, a finite- q phonon emitting a near- Γ acoustic phonon, exactly the process that builds the two-phonon continuum edge. There the minimal-image leak is $O(1)$ on the natural scale of the tensor (Fig. 3, left).
- *Pairwise potentials are structurally immune.* At $L = 3$ the violating image weights multiply only tensor entries whose three legs span three distinct cells; a two-body (bond) potential populates only two-site entries, so the contraction is identically zero. This is why the $M3$ benchmarks (nearest-neighbour bond toy) never showed ASR artifacts — and why real crystals, whose $\Phi^{(3)}$ generically has three-site entries, will. The validation toy therefore adds a three-body term $g_{3b} s_1^2 s_2$ spanning three cells.

The stochastic estimator confirms the picture quantitatively (Fig. 3, right; three-body-dominated chain, $L_c = 3$, fine mesh 48, perturbation at $q_p = 1/2$): as the acoustic partner $\tilde{q}_2 \rightarrow 0$ the plain and ASR-design vertices vanish linearly while the minimal-image one saturates at a plateau $\approx 7.6\times$ the plain value at the smallest probed $\tilde{q}_2 = 1/48$. The interpolated dynamical matrix keeps its own exact D2 ASR throughout (Tensor2 centering + ASR), so the χ factors and the vertex vanish at the same points.

7.5 Response-level validation: leak versus resolution

The kernel-level ASR test is necessary but not sufficient; the real question is whether it affects the final response. In the three-body toy model, the static Lanczos observable $\omega_{\text{eff}} = \sqrt{-1/[M^{-1}]_{00}}$, three-body toy, $g_{3b}=0.08$, separates two independent errors.

(i) *Under-resolved centering.* At $L_c = 3$, a spread-2 tensor sits exactly *on* the Wigner–Seitz tie boundary: the image assignment is Nyquist-ambiguous, and no compact-support kernel — centered, ASR-projected or otherwise — can resolve it. All oscillating designs then redistribute $O(1)$ tensor weight with wrong phases and fail catastrophically at strong coupling: aggregate renorm error 369.6 cm^{-1} for minimal-image (with 6 modes returning spurious non-negative-definite responses) and $\approx 300 \text{ cm}^{-1}$ for the ASR designs (8 such modes), against a two-seed noise floor of 48.1 and *plain* at 50.6 with no failures. This is a centering-resolution failure, not an ASR failure. It says that the coarse cell is too small for any centered interpolation. In that regime the plain tent kernel is the safe fallback.

(ii) *The ASR leak, at resolvable range.* At $L_c = 5$ (tensor interior to the WS cell) the roles invert — near- Γ interpolated modes, renormalization in cm^{-1} against a direct $L_f = 10$ ensemble:

mode	direct	plain	minimal-image	ASR design
$\tilde{q} = 1/10$, acoustic	−56	spurious $g > 0$	−77	−92
$\tilde{q} = 1/10$, optical	−28	−35	−31	−34
$\tilde{q} = 3/10$, acoustic	−413	−234	−396	−407
$\tilde{q} = 3/10$, optical	−793	−341	−733	−752

Here the *plain* kernel is the broken one: the tent bias halves the large renormalizations and flips a near-soft acoustic mode to a spurious non-negative-definite response, while minimal-image and ASR designs both track the reference, the ASR design closest on the strongly renormalized modes — with the qualitative difference that its vertex is exactly ASR-clean, whereas the minimal-image plateau of Fig. 3 is structural and gains weight as the fine mesh densifies the Γ neighbourhood.

The practical rule is therefore:

If the coarse cell resolves the third-order range, use the ASR design. If the range reaches the Wigner–Seitz boundary, no centered compact-support kernel is trustworthy and the plain estimator is the correct fallback.

On the original pairwise toy the ASR design keeps the same precision as the minimal-image windows (Sec. 10).

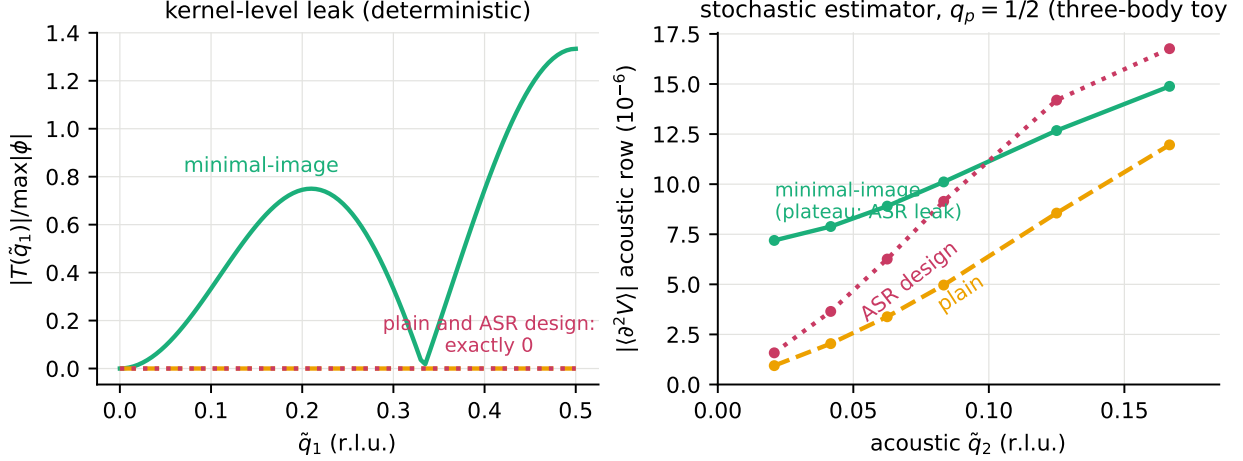


Figure 3: Left: deterministic kernel-level acoustic leak $|T(\tilde{q}_1)|$ (translation contraction of the interpolated three-body test tensor on the $\tilde{q}_2 \rightarrow 0$ leg, normalized to the largest tensor entry). The minimal-image kernel leaks $O(1)$ at generic $\tilde{q}_1$, vanishing only at Γ and at the commensurate point $1/3$; the plain and ASR-design kernels are exactly zero everywhere. Right: the same channel probed with the stochastic estimator ($N_I = 4000$, $q_p = 1/2$): the minimal-image acoustic vertex saturates (ASR leak) while plain and ASR designs vanish with $\tilde{q}_2$.

8 Atom-resolved pinned centering for $L = 2$ real-material cells

The cell-index windows above know only the supercell lattice. They do not see the basis offsets inside the primitive cell. This is fatal for the SnTe benchmark, whose coarse ensemble is only $2 \times 2 \times 2$. In such a cell every nonzero cell difference is a Wigner–Seitz tie at the cell-index level.

The standard tensor route nevertheless succeeds on the same data: `ForceTensor.Tensor3.Center(Far=3)` uses the actual Cartesian interatomic vectors, including the Sn and Te basis positions, and chooses the replica geometry with the smallest triplet perimeter. Thus the information is present in the coarse data; a tensor-free method simply has to use atom-resolved geometry instead of cell-only geometry.

The tensor-free fix is to pin one vertex leg to a primitive atom and coarse origin, then center the other legs relative to that actual atom. For one primitive atom a and one coarse origin $\mathbf{o}$, a perturbation-slot pin is

$$z_{\mathbf{ao}}(b, \mathbf{n}) = \delta_{ab} \delta_{\mathbf{no}}. \quad (15)$$

For a pair-leg atom b in cell-difference class δ , choose the periodic images that minimize the actual Cartesian distance

$$\mathcal{I}_{ab}(\delta) = \arg \min_{\mathbf{m} \in [-F, F]^3} |\boldsymbol{\tau}_b - \boldsymbol{\tau}_a + (\delta + \mathbf{m}\mathbf{L})\mathbf{A}|, \quad (16)$$

with exact ties split uniformly. The same atom-resolved rule is used for the two pair slots. Summing over all coarse origins restores the translation sum removed by the pinned delta; no $1/N_c$ prefactor is applied. The resulting D3 kernel is

$$K_{\text{at}}^{abc}(\delta_1, \delta_2) = \sum_{\mathbf{r}_1 \in \mathcal{I}_{ab}(\delta_1)} \sum_{\mathbf{r}_2 \in \mathcal{I}_{ac}(\delta_2)} \frac{e^{-i\tilde{\mathbf{q}}_1 \cdot \mathbf{r}_1} e^{-i\tilde{\mathbf{q}}_2 \cdot \mathbf{r}_2}}{|\mathcal{I}_{ab}(\delta_1)| |\mathcal{I}_{ac}(\delta_2)|}, \quad (17)$$

with $\mathbf{r} = (\delta + \mathbf{m}\mathbf{L})\mathbf{A}$. At runtime any selected image outside the fundamental supercell is folded back onto the periodic atom and multiplied by the corresponding Bloch phase $\exp[-i\tilde{\mathbf{q}} \cdot (\mathbf{m}\mathbf{L})\mathbf{A}]$; this is just the per- q , per-atom weight path already used by the doubled-support ASR windows. No $\Phi^{(3)}$ value enters Eq. (16).

Table 1: SnTe Γ TO peak positions and ASR quality at near- Γ wavevectors ($|\mathbf{q}| = 0.066 \text{ \AA}^{-1}$ on the 4^3 mesh). All Lanczos runs use $2^3 \rightarrow 4^3$ interpolation with $\sigma = 1.5 \text{ cm}^{-1}$; the tensor-D3 oracle and the direct 4^3 reference employ the deterministic d_3 tensor (centered, $F_{\text{far}} = 3$, ASR). N_I is the number of stochastic configurations; *n. s.* is the number of Lanczos steps. The acoustic D_3 row norm $\|D_3^{\text{ac}}\|$ is the Frobenius norm of the acoustic-band row of the d_2v blocks at $q_{\text{pert}} = \Gamma$; the ASR projector reduces this leakage by a factor of ~ 14000 without shifting the TO peak.

mode	peak (cm^{-1})	static G	N_I	n. s.	passes	$\ D_3^{\text{ac}}\ $
direct 4^3 q-Lanczos	37.56	+ (unstable)	4000	—	—	—
tensor-D3 oracle	37.65	$+5.42 \times 10^6$	4000	250	—	—
atomic_delta (this work)	37.52	$+5.02 \times 10^6$	1000	30	48	3.4×10^{-17}
atomic raw (same windows)	37.52	$+5.14 \times 10^6$	1000	30	48	5.0×10^{-13}
plain window (broken)	51.83	-4.91×10^6	500	30	0	—
old d_3 bubble ($k=20^3$)	38.46	—	—	—	—	—

The Julia D3 estimator is already permutation-symmetric, but that symmetry is expressed as three force-leg channels. One channel has the force on the perturbation slot z ; the other two have the force on w and v . The atom-resolved windows must follow the same split, otherwise they would center the wrong leg. The production path therefore runs three channel families:

$$\text{force on } z : \quad z\text{-pinned windows as in (17),} \quad (18)$$

$$\text{force on } w : \quad w\text{-pinned windows, with } z, v \text{ centered relative to } w, \quad (19)$$

$$\text{force on } v : \quad v\text{-pinned windows, with } z, w \text{ centered relative to } v. \quad (20)$$

When all slot windows are identical, the sum of the three channel calls is algebraically identical to the original all-channel D3 kernel; this is a unit test. For SnTe the pass count is consequently $3 \times 2 \times 8 = 48$ D3 channel passes, but only 32 distinct field sets because many windows are reused.

The strict- Γ ASR follows from the same unit class sums. For every folded atom/cell class, the non-pinned windows have total selected-image weight one; after summing all pinned atoms and origins, the pinned slot has the same property. Thus a uniform acoustic displacement at $q = 0$ sees the same folded sums as the plain commensurate estimator and contracts to zero. Near Γ , however, the first moment of the selected images matters. That is why the final mode uses a conservative q-space acoustic projector (**atomic_delta**) instead of averaging away the pinned atom. A prototype with a uniform pinned slot lost the SnTe basis-offset information and reverted to the bad plain peak near 52 cm^{-1} .

Before using the construction in production, we performed an oracle test. The periodic SnTe $\Phi^{(3)}$ tensor is contracted with candidate geometry kernels, but only to measure the geometry error; the production estimator does not use this tensor. On the interpolated Γ -pair blocks of the failed $2^3 \rightarrow 4^3$ case, the plain cell-index kernel has mean relative Frobenius error 1.048 (max 1.582) against **Tensor3.Center(Far=3)**. The leg-wise atomic kernel has mean 0.008 (max 0.011), while the full perimeter target is zero within numerical precision. Thus the cheap leg-wise atomic kernel is already close enough to the validated tensor oracle.

The SnTe spectral-function benchmark confirms this. The interpolated Γ TO spectrum (D3-only, no LO-TO splitting) is shown in Fig. 4 and the peaks are summarised in Table 1.

The key result is that **atomic_delta** keeps the correct TO peak (37.52 vs 37.56 cm^{-1} for the direct 4^3 reference) while the q-space acoustic projector dramatically improves the near- Γ acoustic sum rule. The acoustic-mode D_3 vertex norm at the smallest non-zero q on the 4^3 mesh ($|\mathbf{q}| \approx 0.066 \text{ \AA}^{-1}$) is reduced from 5×10^{-13} (raw atomic) to 3.4×10^{-17} under **atomic_delta** – a factor of ~ 14000 improvement (Fig. 5). At exact Γ both variants give machine zero because the folded-class-sum property of the permutation-correct atomic windows already enforces the hard partition.

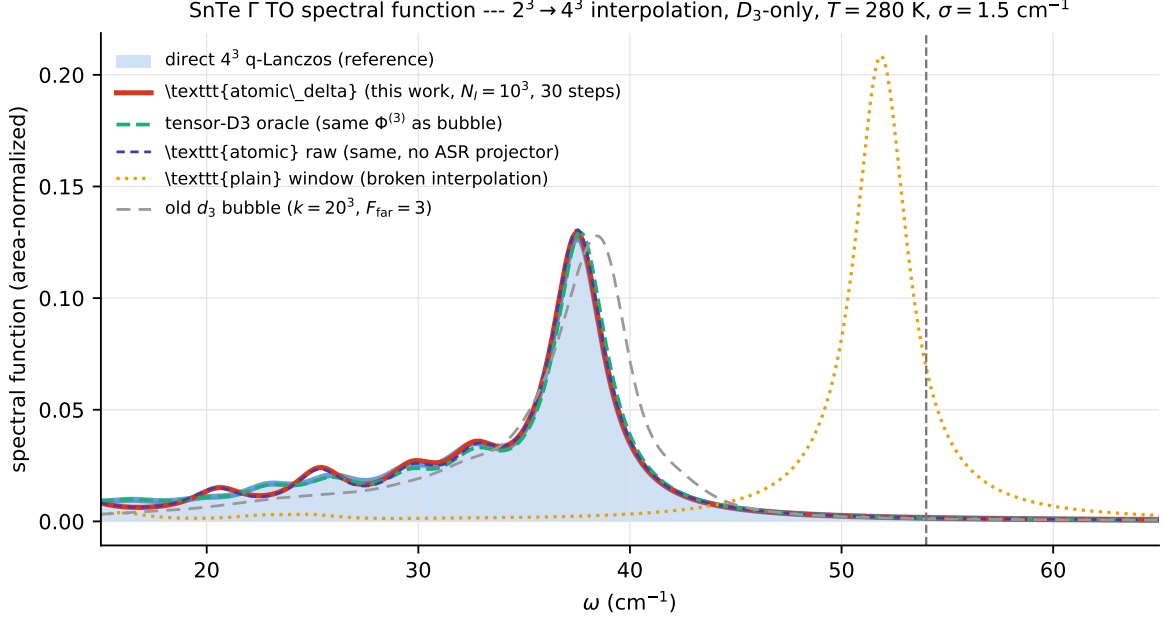


Figure 4: SnTe Γ TO spectral function, $2^3 \rightarrow 4^3$ interpolation at $T = 280$ K with 1.5 cm^{-1} smearing, D3-only (no D4). The `atomic_delta` curve (red) uses the same 48-pass permutation-correct atom-resolved windows as the raw `atomic` mode (violet) plus a conservative q -space acoustic projector on the windowed Bloch fields. The tensor-D3 oracle (green dashed) and the direct 4^3 q -space Lanczos (blue shaded) are reference calculations; the old d_3 dynamic bubble (grey, $k = 20^3$, $F_{\text{far}} = 3$) is also shown. The `plain` window curve (yellow dotted) illustrates the broken $L = 2$ interpolation without atom-resolved centering.

The static response has $g > 0$ (unstable) for `atomic_delta`, `atomic`, and the tensor-D3 oracle, matching the direct 4^3 calculation; the broken `plain` interpolation instead produces a spurious stable response ($g < 0$, $w_{\text{eff}} = 49.5 \text{ cm}^{-1}$). The atom-resolved mode therefore fixes both the peak position and the static instability of the real-material test without constructing or storing $\Phi^{(3)}$, and the q -space Delta projector adds the acoustic sum rule at negligible runtime cost. Its current cost is the number of force-channel/pinned-atom/origin passes (SnTe: $3 \times 2 \times 8 = 48$ D3 pass calls). This makes `atomic_delta` the recommended tensor-free D3 mode for underresolved small supercells where cell-index windows cannot resolve atomic-basis ties.

9 Implementation and cost

The implementation (branch `qspace_interpolation`) consists of `Modules/QSpaceInterpolation.py` and `tdscha.qspace.jl`. The Python class `QSpaceLanczosInterp` performs the fine-mesh setup: it builds the fine q list, hashes q points for $O(1)$ lookup, constructs the $O(N_f)$ pair map, interpolates the SSCHA dynamical matrix, builds the windowed Bloch fields, and dispatches the Lanczos action.

The Julia side has two equivalent slot kernels. The scalar kernel is useful for readability and tests. The batched kernel groups configurations at a fixed symmetry operation: symmetry rotations become sparse-dense products, and the $\alpha^{(1)}$ intermediates and two-phonon accumulations become GEMMs. The scalar and batched kernels agree to 10^{-20} .

For a window design with R runtime passes, one application of the anharmonic part scales as

$$O(R N_{\text{conf}} n_{\text{sym}} N_f n_b^2).$$

For the separable minimal-image designs, $R = K^3$. For the plain estimator, $R = 1$. For the SnTe atom-resolved D3 mode of Sec. 8, $R = 3n_{\text{at}}N_c$ because there are three force channels, each

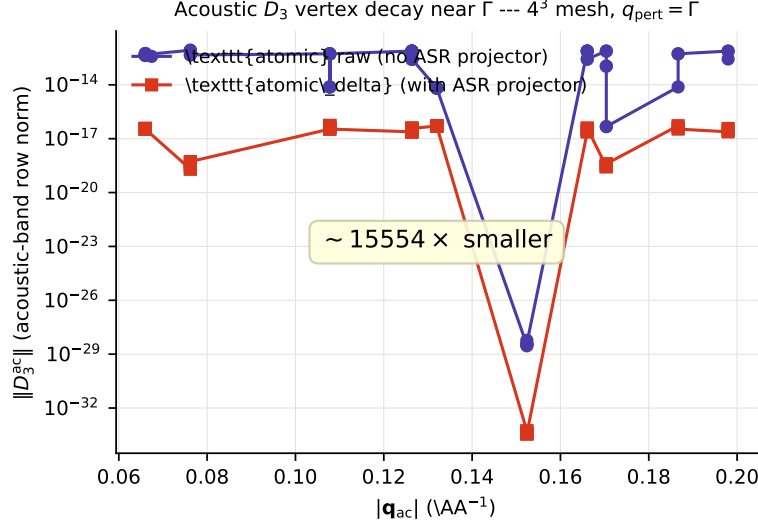


Figure 5: Acoustic D_3 vertex norm $\|D_3^{ac}\|$ (Frobenius norm of the acoustic-band row of the d_2v block at $q_{\text{pert}} = \Gamma$) versus the acoustic-leg wavevector $|q_{ac}|$ on the 4^3 mesh. The `atomic_delta` projector (red squares) reduces the near- Γ acoustic leakage by about four orders of magnitude compared to the raw `atomic` mode (violet circles), while both modes give the same TO peak position (37.52 cm^{-1}).

pinned to one primitive atom and all coarse origins; in the 2^3 SnTe cell this is $3 \times 2 \times 8 = 48$ D3 passes. In all cases R is fixed by the coarse cell and the chosen kernel, not by the fine mesh, so the scaling in N_f is linear. Figure 6 verifies this over $N_f = 8 \dots 128$.

10 Benchmarks

The first validation problem is an anharmonic diatomic chain. It is small enough that we can generate both a coarse ensemble and a direct fine ensemble for the same model, so interpolation can be tested against a true fine-supercell reference. The model has harmonic springs K_1, K_2 and a bond cubic anharmonicity $g_3 s^3/3$; the weak-coupling tests use $g_3 = 0.1$, giving $5 \dots 40 \text{ cm}^{-1}$ renormalizations on $\sim 3000 \text{ cm}^{-1}$ modes. The cell is triclinic with symmetry $P1$, so point-group symmetries cannot hide sign, phase, or conjugation mistakes. Some test wavevectors are not time-reversal invariant ($\mathbf{q} \neq -\mathbf{q}$), which also checks the complex gauge conventions.

Exactness checks. These tests do not rely on physics judgment; they check identities that must hold exactly.

1. If the fine mesh equals the coarse mesh, the interpolated Lanczos coefficients reproduce the production `QSpaceLanczos` coefficients to 10^{-15} , both with and without pre-filtering.
2. The interpolated dynamical matrix reproduces the direct larger-supercell dispersion exactly for this range-1 harmonic model.
3. Rigid displacement shifts and constant force offsets leave the fields invariant.
4. The batched and scalar Julia kernels agree to machine precision, and the windowed path with trivial windows equals the plain path identically.

Physics: interpolation vs. direct fine supercell. The decisive comparison is between a coarse $(1, 1, 3)$ ensemble interpolated to the $(1, 1, 6)$ mesh and a direct $(1, 1, 6)$ ensemble of the same model. The error metric is the anharmonic renormalization $\omega_{\text{eff}} - \omega_{\text{SSCHA}}$, not the absolute

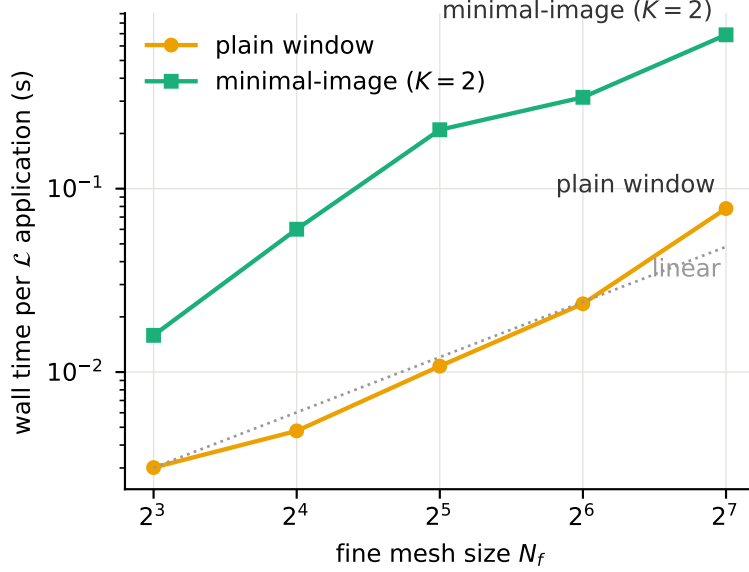


Figure 6: Wall time of one application of $\mathcal{L}$ versus the fine-mesh size N_f (chain toy, $N_I = 1000$ configurations, coarse $L_c = 4$): plain estimator and minimal-image windows ($K=2$, i.e. five kernel passes). Both follow the ideal linear trend (dotted).

frequency, because the harmonic part is already accurately interpolated and would otherwise dominate the comparison. Table 2 and Fig. 7 summarize the aggregate error $\sum |\Delta_{\text{int}} - \Delta_{\text{dir}}| / \sum |\Delta_{\text{dir}}|$:

	aggregate error	comment
noise floor (two direct seeds)	0.103	statistical reference
plain window	0.299	tent-kernel bias
minimal-image windows, 1 origin	0.158	
minimal-image windows, 3 origins	0.122	at the noise floor
ASR design, 3 origins (Sec. 7)	0.150	exact ASR, no precision lost
ASR design + decay weight $\rho=0.4$	0.196	
plain, <i>no</i> pre-filter	$\approx 2\times$ deficit	systematic (Sec. 6)
plain, <i>no</i> s_3, s_4 rescaling	$\approx 2\times$ overshoot	systematic (Sec. 5)

Table 2: Aggregate renormalization error of the interpolated Lanczos against direct fine-supercell calculations (chain toy, $L_c = 3 \rightarrow L_f = 6$, $N_I = 4000$, all 33 valid modes). The designed windows remove essentially all resolvable interpolation error; the last two rows document the failure modes that the pre-filter and the vertex rescaling correct.

11 Example application: spectral functions at interpolated q and the standard d_3 bubble

As an end-to-end check, we compute one-phonon spectral functions at wavevectors absent from the coarse mesh and compare them with the standard d_3 dynamic bubble from `cellconstructor.Spectral.get_dia`. In this toy model the analytic $\Phi^{(3)}$ is known, so the bubble route can use the exact centered and ASR-projected tensor. The Lanczos calculation disables D_4 so that both methods contain the same diagrammatic content. Because the model is weakly coupled, the perturbative bubble and the resummed TDSCHA Lanczos should agree. They do: Fig. 8 shows agreement at the energy-grid resolution, about 2.3 cm^{-1} , on shifts of $15 \dots 40 \text{ cm}^{-1}$.

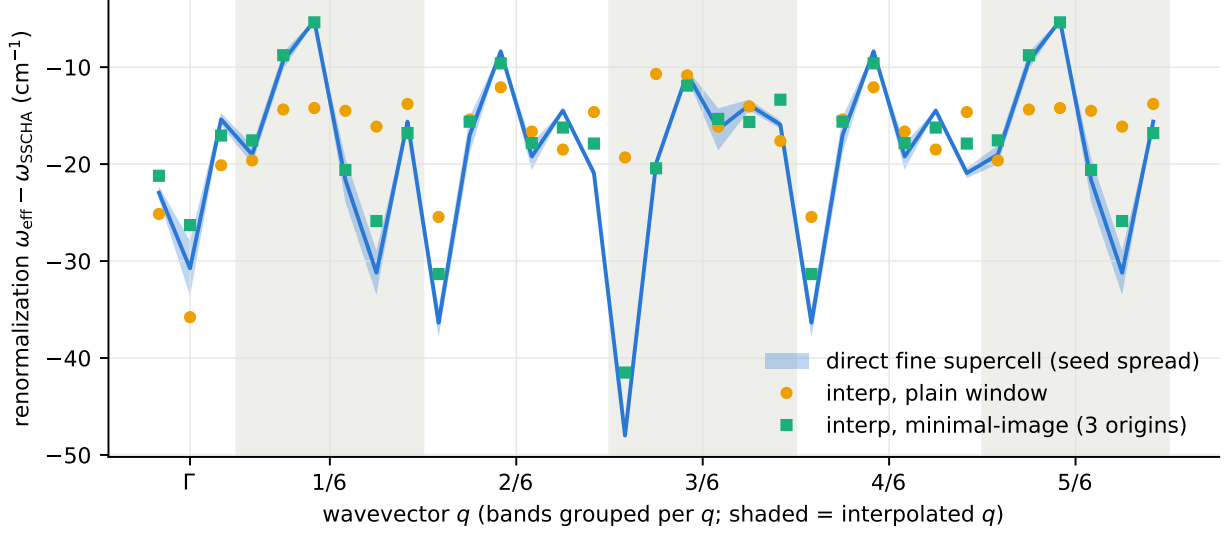


Figure 7: Mode-resolved anharmonic renormalization on the fine (1,1,6) mesh: direct fine-supercell reference (two independent seeds indicate the stochastic spread) versus the interpolated Lanczos from the coarse (1,1,3) ensemble with the plain window and with the designed minimal-image windows (3 origins). Vertical guides mark the interpolated (non-commensurate) wavevectors, including the non-TRI pair $q = \pm 1/6$.

q	band	ω_{SSCHA} (cm^{-1})	peak (Lanczos interp) (cm^{-1})	peak (d_3 bubble) (cm^{-1})
2/16	4	2205.3	2185.6	2183.2
2/16	5	2250.6	2214.2	2216.6
6/16	4	2013.4	1994.9	1992.5
6/16	5	2058.5	2040.2	2040.2

12 Fourth order: spectral functions where D_4 is fundamental

The bubble benchmark of Sec. 11 validates the *cubic* interpolation against an analytic diagram, with D_4 switched off. The four-phonon vertex enters through four-field Bloch averages. These averages are evaluated off-grid by the same stochastic machinery and are rescaled by $s_4 = N_c/N_f$ (Sec. 5). The initial implementation used a single plain-window pass for D_4 , based on the expectation that fourth-order correlations are shorter-ranged and smaller. This section separates what is already validated from what is still open: the plain pass works on the controlled chain benchmark, but it is not sufficient for underresolved SnTe cells, where a genuine four-leg D4 centering is needed.

There is no simple analytic diagram to compare against, because the static one-phonon self-energy of the *bare* quartic vertex — the tadpole $\frac{1}{2} \sum_k d_4(q\nu, -q\nu, k\mu, -k\mu) \chi_{k\mu}$ — is exactly the quantity SSCHA already resums into the auxiliary frequency ω_{SSCHA} . A *purely* quartic model therefore produces *zero* anharmonic renormalization of the one-phonon line: with $D_3 = 0$ the initial single-phonon state never couples to the two-phonon sector, the Lanczos chain terminates at first step, and $\omega_{\text{eff}} = \omega_{\text{SSCHA}}$ identically (we verified this numerically to machine zero). What survives of D_4 in the TDSCHA response is its action *inside* the two-phonon continuum that D_3 opens; it is genuinely a fourth-order effect, but only visible together with a nonzero third order.

The controlled validation uses internal consistency in a mixed cubic-quartic regime. The anharmonic diatomic chain is given a moderate cubic bond term ($g_3 = 0.15$) and a *strong* quartic one ($g_4 = 2.0$), so that the quartic vertex reshapes the spectrum by several tens of cm^{-1} . We com-

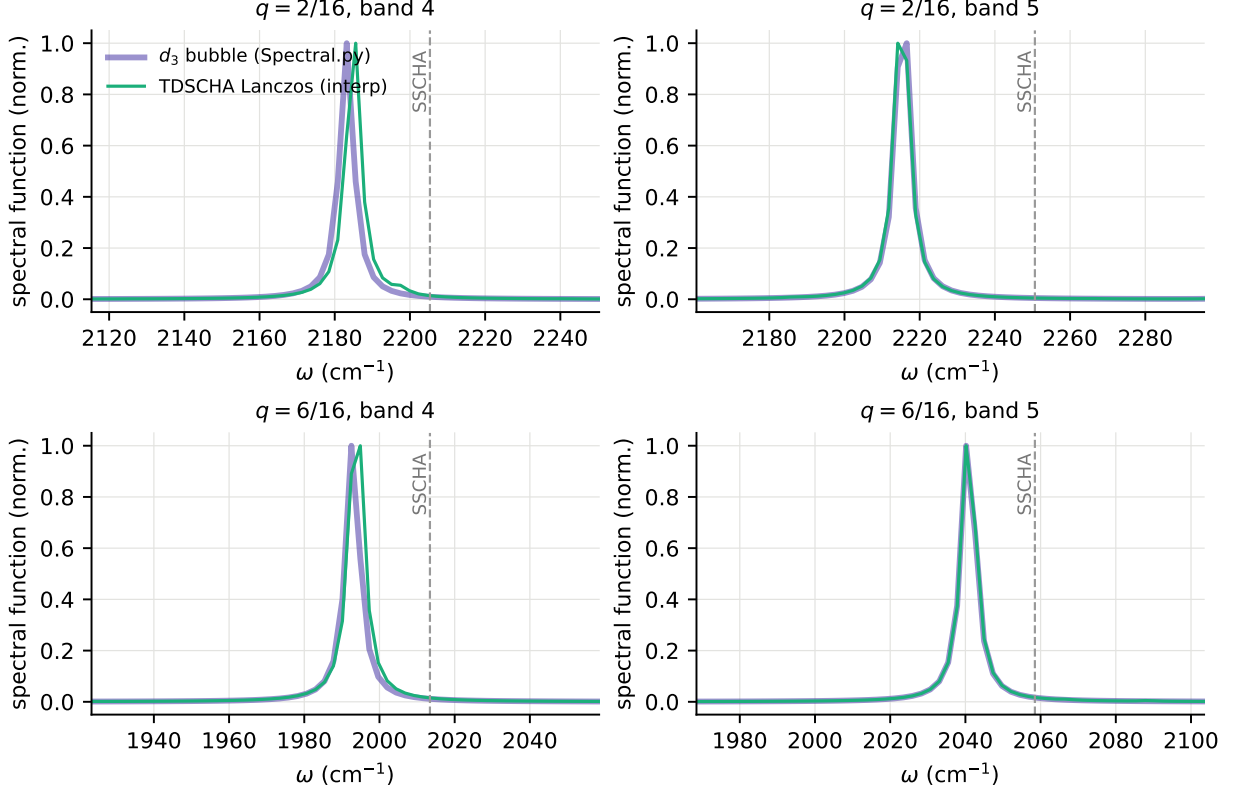


Figure 8: One-phonon spectral functions at interpolated wavevectors ($q = 2/16$ and $6/16$ along the chain, two optical bands): fine-mesh TDSCHA Lanczos interpolated from the coarse $(1,1,4)$ ensemble (minimal-image windows) versus the standard d_3 dynamic bubble of `cellconstructor.Spectral` with the exact centered $\Phi^{(3)}$ tensor on the same k -mesh and smearing. The SSCHA frequency is marked.

pute the full D_3+D_4 one-phonon spectral function $A(q, \omega) = -\frac{1}{\pi} \text{Im } G(q, \omega)$ (continued-fraction Green function, $N_{\text{steps}} = 110$, smearing $\approx 6.6 \text{ cm}^{-1}$) at wavevectors of the fine $(1, 1, 9)$ mesh, in two ways: interpolated from a coarse $(1, 1, 3)$ ensemble (minimal-image windows, two origins) and directly from an independent $(1, 1, 9)$ ensemble of the *same* chain ($N_I = 6000$ each). As a control we also record each spectrum with D_4 switched off.

Figure 9 and the table below show the outcome. In this model the plain D_4 pass is enough: the quartic vertex hardens the peak by $30 \dots 56 \text{ cm}^{-1}$ relative to the D_3 -only line (dashed), a shift comparable to the SSCHA frequency’s own anharmonic budget and far larger than any interpolation error. The interpolated full spectrum (aqua) nonetheless lands on the direct fine-mesh reference (blue) to within one energy-grid step, $\lesssim 2 \text{ cm}^{-1}$ on the peak and $L_1 \approx 0.15 \dots 0.22$ on the whole normalized line shape (with $L_1 = \int |\hat{A}_{\text{interp}} - \hat{A}_{\text{direct}}| d\omega$, $\hat{A}$ normalized to unit area). Dropping D_4 from the interpolation raises L_1 to $\approx 1.7 \dots 1.8$, so the line shapes are nearly disjoint. This proves that the stochastic scaling and q -space placement of the D_4 terms are correct in a resolved benchmark. It does not prove that the plain image assignment is adequate for every real material.

q	band	ω_{SSCHA} (cm^{-1})	peak (cm^{-1})		D_4 shift (cm^{-1})	interp–dir (cm^{-1})	L_1 full / D_3
			D_3	D_3+D_4			
1/9	5	2258.0	2198.8	2237.8	+39.0	+1.9	0.22/1.81
2/9	5	2177.4	2104.0	2159.8	+55.8	0.0	0.15/1.80
4/9	4	1951.4	1907.0	1936.7	+29.7	−1.9	0.17/1.70

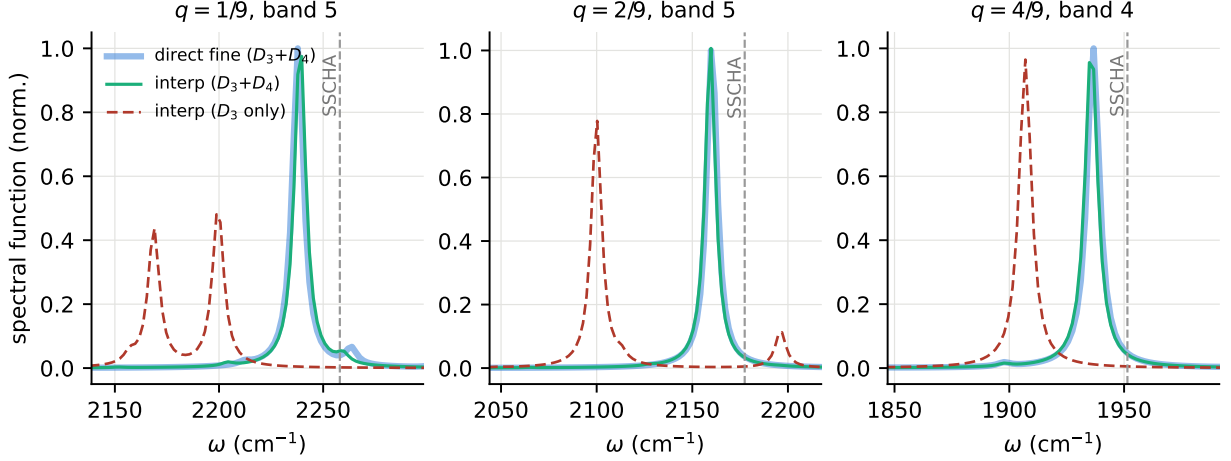


Figure 9: One-phonon spectral functions in a D_4 -dominated regime ($g_3 = 0.15$, $g_4 = 2.0$) at interpolated wavevectors of the $(1, 1, 9)$ mesh. Blue: direct fine-mesh TDSCHA Lanczos (D_3+D_4). Aqua: interpolated from the coarse $(1, 1, 3)$ ensemble (D_3+D_4) — it overlays the direct reference. Red dashed: the same interpolation with the fourth order dropped (D_3 only), displaced by the quartic hardening of $30 \dots 56 \text{ cm}^{-1}$. The SSCHA frequency is marked; curves normalized to the direct peak.

Peaks are the direct fine-mesh values; the “ D_4 shift” is D_3+D_4 minus D_3 -only (direct), and “interp–dir” is the interpolated D_3+D_4 peak minus the direct one. At the operator level the two design choices are exact rather than merely accurate: the D_4 block of $\mathcal{L}$ scales strictly linearly in s_4 and is bit-identical across the plain, minimal-image and ASR window designs (it is produced by the same plain-window pass in all three), which the test suite checks directly.

12.1 Atomic centering of the four-phonon vertex

The SnTe tests show where the simple D_4 treatment fails. In the underresolved $L = 2$ real-material cells of Sec. 8, the plain-window D_4 pass has the same basis-tie ambiguity that broke the plain D_3 estimator. A low-symmetry benchmark on the SnTe force field (R3m phase at $T = 180 \text{ K}$, where the free-energy Hessian requires the fourth order: the A_1 order-parameter mode is $76.2 \rightarrow 25.3 \text{ cm}^{-1}$ with the D_3 bubble alone and $\rightarrow 39.1 \text{ cm}^{-1}$ with D_4) showed that the $2 \times 2 \times 2 \rightarrow 4 \times 4 \times 4$ interpolation with plain D_4 systematically *under-captures* the D_4 renormalization of the spectral first moment: at $N = 4000$ the direct fine-mesh D_4 shift of $\langle \omega \rangle$ is -22.9 cm^{-1} but the interpolation recovers only -14.8 cm^{-1} (64%), while the D_3 -only moments agree to $\sim 5 \text{ cm}^{-1}$ — the deficit is D_4 -specific.

The difficulty is structural. The mathematical vertex has four logical legs, $\Phi^{(4)}(E_w, E_v, I_w, I_v)$: two external two-phonon legs and two internal legs contracted with the two-phonon vector. The existing Julia estimator stores only two pair-slot field families, w and v , and reuses them for both internal and external legs. Any D_4 centering that manipulates only these two slots is therefore a projection of a four-leg geometry onto a two-slot implementation.

Two such projections were tested. The first is the **reference** mode. It runs the D_4 term on the channel-0 atomic passes: all four D_4 legs carry the same atom-resolved window $W_{a,o}$ relative to an external pinned atom. This mode is tensor-free, permutation-symmetric, ASR-safe under the existing q -space projector, and exactly equal to plain D_4 at commensurate q . But the pinned atom is not one of the four D_4 vertex legs, so the centering reference is physically wrong.

What the external-reference scheme does *not* get right is the centering reference: the pinned atom is not one of the four vertex legs, so all four legs are imaged relative to an external point and every leg suffers the $L=2$ tie dilution, compounded over the four-fold product. Consistently, on the R3m benchmark the centered D_4 vertex block differs from the plain one by 78% in norm, yet

the A_1 observable barely moves (-5.6 vs -6.1 cm^{-1} , both far from the direct -16.1 at $N = 1000$).

The second projection is `d4_center="leg"`. It is the direct analogue of the D3 force-channel construction: pin the D4 leg that carries the force. Across the estimator terms the force leg occupies all four positions: the two external legs through the $y_w(q_1) f_Y x_v(q_2)$ and $f_Y x_w(q_1) y_v(q_2)$ parts of the $\langle x x \alpha \rangle$ -type term, and the two internal ones through the f_ψ -contracted term ($\overline{\alpha_1 x_v^*} f_\psi y_w^*$ and its $w \leftrightarrow v$ mirror). Splitting the kernel by these four force positions (channel gates `d4_force_ew/ev/iw/iv`, the D_4 analogue of the D_3 1/3+2/3 split) reveals a decisive simplification: within each channel the pinned delta touches only the *force* (Y) field of one slot, while the displacement (X) field of the same slot always belongs to the other, windowed legs. The centering is therefore realized without any new field family, by passing per slot the X field of the atomic window $W_{a,o}$ and the Y field of the pinned delta: one W -force call (channels $ew+iw$, $Y_w = \delta$) and one V -force call ($ev+iv$, $Y_v = \delta$) per pinned atom/origin, i.e. $2 n_{\text{at}} N_c$ extra kernel calls and zero new field sets. Because every gated term is linear in the delta-carrying Y field, the sum over pinned atoms and full coarse origins reconstructs the force-leg atom sum with no prefactor, and at commensurate q it collapses to the plain D_4 estimator configuration by configuration. The summed four-channel kernel $\sum_\ell \delta(\ell) W \otimes W \otimes W$ is symmetric under leg exchange, so the 4-leg permutation symmetry of $\Phi^{(4)}$ is preserved exactly, as for D_3 ; the channel split and the commensurate identity are locked by unit tests (`test_d4_center.py`).

This construction passes the algebraic tests: the channel split reproduces the all-channel D4 call, the commensurate identity holds, and the summed four-channel kernel is formally symmetric. But the SnTe spectra show that these identities are necessary, not sufficient. The first complete Fm-3m smoke benchmark with all three D_4 modes ($T = 280$ K, quartic couplings scaled by 3, $N = 300$, 25 Lanczos steps) gives a direct spectral-moment D_4 shift of -7.25 cm^{-1} . Plain D_4 captures only -1.65 cm^{-1} (23%), the external-reference centering improves to -3.63 cm^{-1} (50%), but the pin-force-leg construction gives the *wrong sign*, $+2.05$ cm^{-1} (-28% capture) and the worst line-shape distance. A larger, cleaner partially completed run ($T = 280$ K, $p_4 \times 2$, $p_3 \times 1.6$, $N = 2000$, first off-grid shell) confirms that the residual error is mode dependent rather than a scalar normalization: for band 5, plain D_4 captures $+5.93/+8.71 = 68\%$ of the direct first-moment shift, whereas for band 0 it slightly overshoots, $+1.48/+1.33 = 111\%$. At the physical $T = 400$ K couplings the total shift is too small to be decisive, but the completed part of the run again shows only partial capture (plain 14%, reference 41%). The current conclusion is therefore negative: the implemented `d4_center="leg"` split preserves channel sums and the commensurate limit, but it does not yet realize the physically correct four-leg minimal-image interpolation. The next diagnostic must compare the four force channels at the operator level against an explicit geometry-only four-leg target before using longer production spectra as evidence. The geometry diagnostic makes the failure concrete. A single w or v window simultaneously represents an external leg and an internal leg at different momenta. Pinning only the force field does not independently center the other three D4 legs. As a safety patch, the boolean production switch `d4_center=True` has been changed back to the legacy external-reference mode; the failed pin-force-leg construction remains available only as the explicit diagnostic option `d4_center="leg"`. This avoids silently selecting the wrong-sign scheme while the true four-leg centering target is unresolved. The patch is verified in the SnTe environment by running with the branch overlay `PYTHONPATH=/tmp/tdscha_pyshim`: a D4-only operator application at the first off-grid shell gives relative difference 0.0 between `d4_center=True` and `d4_center="reference"`. Without this overlay, `micromamba run -n sscha` imports the installed site-package copy of TDSCHA and does not exercise the working tree. A pure geometry scan on the SnTe $L = 2$ cell gives the immediate reason: comparing the current force-star image assignment with a true four-leg perimeter target over all 8192 atom/cell classes (nearest image shell) gives total-variation distance mean 0.614, maximum 1.0, and nonzero mismatch in 85.3% of cases. Thus the problem is not the q -space ASR projector (which gives the same behavior with `atomic` and `atomic_delta`), but the geometry target itself: the force-star product is not a faithful four-leg centering rule. The word “reference” in `d4_center="reference"` should therefore not be read as a stored- $\Phi^{(4)}$ oracle. That mode

remains tensor-free: it reuses the channel-0 atom/origin stochastic passes and applies the same atom-resolved window to the D4 legs relative to an external pinned atom, averaging over pinned atoms and origins. The pinned atom is not one of the four D4 vertex legs, so this is a legacy comparison/guardrail rather than the correct four-leg centering. In the branch-overlay SnTe smoke benchmark (PYTHONPATH=/tmp/tdscha_pyshim, $N = 300$, 25 Lanczos steps, quartic scale 3), the direct D4 first-moment shift is -7.25 cm^{-1} while reference mode gives -3.63 cm^{-1} (50% capture, normalized $L^1 = 0.391$ against the direct $D_3 + D_4$ spectrum).

Restarting from a four-leg target. The replacement design is to lift the stochastic D_4 interpolation from the two reused pair slots to the four logical legs of the two-phonon-to-two-phonon vertex: the external legs (E_w, E_v) and the internal legs (I_w, I_v) contracted with the two-phonon vector. On the atom-resolved extended lattice one first builds a purely geometric target $T_4(E_w, E_v, I_w, I_v)$ by assigning the periodic images of the four atoms that minimize a symmetric cluster cost, for example the complete-graph sum of squared pair distances, with exact tie splitting. This target is then projected onto the linear subspace that enforces the three invariants required of the estimator: the 24 four-leg permutations, unit folded class sums on every commensurate image class (so the coarse-grid operator is exactly the current plain D_4 operator), and translational acoustic sum rules on every leg. This step still uses only geometry and linear algebra; no entry of $\Phi^{(4)}$ is formed, stored or fitted.

The projected target is not meant to be evaluated as a dense four-index object. Instead it should be factorized once into a small separable sum

$$T_4^{\text{ASR}} \simeq \sum_{r=1}^{R_4} c_r A_r(E_w) B_r(E_v) C_r(I_w) D_r(I_v), \quad (21)$$

with atom-resolved one-leg factors whose folded class sums satisfy the same partition and ASR constraints. The runtime implementation is then a slot-lifted D_4 kernel, selected by a new explicit mode such as `d4_center="factor4"`, which accepts four logical D4 field slots rather than reusing the same w, v arrays for both internal and external legs. Each rank term is one batched D4 pass over the existing $O(N_f)$ pair list, so the cost is $O(R_4 N_{\text{conf}} N_{\text{sym}} N_f n_b^2)$ and remains linear in the interpolation mesh as long as the factor rank R_4 is fixed by the coarse-cell geometry rather than by N_f . The validation gate is ordered accordingly: pure SnTe $L = 2$ geometry (where the old force-star target must fail), exact commensurate operator identity, Hermiticity, acoustic-leg scaling, toy-model rank convergence, and only then the Fm- $\bar{3}m$ first-off-grid-shell SnTe benchmark with the plain, reference and leg modes kept as negative controls.

Equivalence: this is not a low-rank fit of $\Phi^{(4)}$. The object being factorized in Eq. (21) is the *image-assignment kernel*, not the fourth-order force constant. The proof is the same kernel argument as Sec. 3.2, now with four logical slots. Collapse atom and Cartesian indices into a and let $i = (a, \mathbf{R})$ denote an atom in one folded coarse supercell. For one ordered D_4 channel take four logical fields $z_\ell(i)$, with one of them the force field y and the other three displacement fields x (the four force positions are summed afterward). A rank- r windowed field is

$$Z_{\ell r}(\mathbf{q}_\ell) = \frac{1}{\sqrt{N_c}} \sum_i W_{\ell r}(i; \mathbf{q}_\ell) e^{-i\mathbf{q}_\ell \cdot \mathbf{R}_i} z_\ell(i), \quad \ell = 1, \dots, 4. \quad (22)$$

The rank- r contribution to the stochastic estimator is the same four-field product already evaluated by the q-space Lanczos kernel,

$$\hat{C}_4^{(r)}(\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \mathbf{q}_4) = \langle Z_{1r}(\mathbf{q}_1) Z_{2r}(\mathbf{q}_2) Z_{3r}(\mathbf{q}_3) Z_{4r}(\mathbf{q}_4) \rangle. \quad (23)$$

By multilinearity,

$$\sum_r c_r \hat{C}_4^{(r)} = \frac{1}{N_c^2} \sum_{i_1 i_2 i_3 i_4} K_R(i_1, i_2, i_3, i_4; \mathbf{q}_1, \dots, \mathbf{q}_4) e^{-i \sum_\ell \mathbf{q}_\ell \cdot \mathbf{R}_{i_\ell}} \langle z_1(i_1) z_2(i_2) z_3(i_3) z_4(i_4) \rangle, \quad (24)$$

where

$$K_R = \sum_r c_r W_{1r}(i_1; \mathbf{q}_1) W_{2r}(i_2; \mathbf{q}_2) W_{3r}(i_3; \mathbf{q}_3) W_{4r}(i_4; \mathbf{q}_4) \quad (25)$$

is the factorized geometry kernel. Thus the factorization changes only the real-space image weights multiplying the same stochastic four-field average. It is not a decomposition, compression, or fit of $\Phi^{(4)}$.

At SSCHA stationarity, Gaussian integration by parts gives for a force leg in slot 4

$$\langle x_1 x_2 x_3 y_4 \rangle = -G_1 G_2 G_3 \left\langle \frac{\partial^4 V}{\partial u_1 \partial u_2 \partial u_3 \partial u_4} \right\rangle, \quad (26)$$

up to the disconnected pieces proportional to $\langle u \otimes \delta f \rangle$, which vanish when the SSCHA auxiliary matrix is converged. The pre-filtering factors of Sec. 6 cancel the propagators G_ℓ exactly as in the D_3 derivation. Therefore Eq. (24) is, in the infinite-sampling limit, precisely the q-space contraction of the fourth-order vertex interpolated with the kernel K_R . If $K_R = T_4^{\text{ASR}}$ exactly, the stochastic estimator is equivalent to forming a centered and ASR-projected $\Phi^{(4)}$ with that same geometry target and contracting it with the two-phonon vector; the implementation simply performs the contraction before any tensor is ever materialized. If the factor rank is finite, the only deterministic error is the kernel error $K_R - T_4^{\text{ASR}}$, not an uncontrolled fit to force constants.

Exact identity at commensurate $\mathbf{q}$. The commensurate proof is independent of the force field. Let $[i]$ denote an atom/cell class modulo the coarse supercell, and let $i + nL$ denote any extended image of that class. For commensurate wavevectors $\mathbf{q}_\ell$ of the coarse mesh, $e^{-i\mathbf{q}_\ell \cdot (\mathbf{R}_i + nL)} = e^{-i\mathbf{q}_\ell \cdot \mathbf{R}_i}$, and the periodic configuration has $z_\ell(i + nL) = z_\ell(i)$. If the projected kernel satisfies the partition constraint

$$\sum_{n_1 n_2 n_3 n_4} K_R(i_1 + n_1 L, i_2 + n_2 L, i_3 + n_3 L, i_4 + n_4 L) = 1 \quad (27)$$

for every folded quadruple $([i_1], [i_2], [i_3], [i_4])$, then regrouping Eq. (24) by folded classes gives

$$\begin{aligned} \sum_{\{n_\ell\}} K_R(i_1 + n_1 L, \dots, i_4 + n_4 L) e^{-i \sum_\ell \mathbf{q}_\ell \cdot (\mathbf{R}_{i_\ell} + n_\ell L)} \prod_\ell z_\ell(i_\ell + n_\ell L) \\ = e^{-i \sum_\ell \mathbf{q}_\ell \cdot \mathbf{R}_{i_\ell}} \prod_\ell z_\ell(i_\ell), \end{aligned} \quad (28)$$

which is exactly the unwindowed coarse-supercell stochastic product. This holds configuration by configuration and for arbitrary displacements, forces and Lanczos vector α_1 ; no ensemble average and no properties of $\Phi^{(4)}$ are used. Consequently any factorized four-leg window satisfying Eq. (27) reproduces the current plain D_4 q-space operator exactly at all commensurate wavevectors. The ASR and permutation projections can change the off-grid image assignment only inside the null space of this partition constraint.

Tensor-free fitting of the kernel. The fit of Eq. (21) can also be performed without building a dense fourth-rank target array. What is needed by constrained ALS or a randomized CP fit are contractions of the target with three one-leg vectors, for example

$$\mathcal{F}_T[p, q, r](i_1) = \sum_{i_2 i_3 i_4} T_4^{\text{ASR}}(i_1, i_2, i_3, i_4) p(i_2) q(i_3) r(i_4). \quad (29)$$

This is exactly the geometry analogue of contracting three displacement legs of $\Phi^{(4)}$ to obtain a force-like object on the remaining leg, but the contraction uses only the image-assignment oracle T_4 and the linear ASR/partition/permutation projectors. The oracle returns the minimal four-leg image weights and tie splits for a requested quadruple; the projected contraction can be applied

matrix-free by composing this oracle with the constraint projectors. The ALS normal equations, or a randomized objective based on probes $\langle a \otimes b \otimes c \otimes d, T_4^{\text{ASR}} - K_R \rangle$, therefore require only target contractions such as Eq. (29). Neither the force-constant tensor $\Phi^{(4)}$ nor even a dense stored T_4 is required. The stored runtime data are only the one-leg factors and coefficients entering Eq. (21).

Choosing a clean SnTe benchmark. The R3m SnTe benchmark is physically relevant but technically awkward: the interpolated fine mesh contains imaginary modes, and masking them can obscure the D4 interpolation error. The quantitative benchmark was therefore moved to the cubic Fm-3m phase, where the auxiliary dynamical matrix interpolates cleanly from 2^3 to 4^3 and the q-space Lanczos does not need symmetry post-processing.

The D4 signal must be made large enough to measure, but this cannot be done by simply multiplying the quartic force constants. If the quartic couplings are scaled and the SSCHA minimum is recomputed, the quartic tadpole is resummed into the auxiliary dynamical matrix: the Γ TO auxiliary frequency hardens from 54 to 86 cm^{-1} , fluctuations are suppressed, and the residual D4 Hessian shift shrinks to +1.1 cm^{-1} . A large D4 signal obtained without re-minimizing is therefore an artifact of violating the SSCHA stationarity condition.

The useful knob is the cubic coupling. In the cubic phase, parity makes the SSCHA minimum independent of d_3 , so d_3 can be scanned without re-minimization. The D4 response diagrams visible in the two-phonon sector scale as $d_3^2 d_4$, while the auxiliary loop does not carry d_3 . At the $2p_4$ minimum the measured scan gives $\Sigma_{\text{bubble}} = 1883 s^2$ and $\Sigma_{D_4} = 172 s^2 \text{cm}^{-2}$ for $p_3 \rightarrow s p_3$. The benchmark therefore uses the first off-grid shell of the 4^3 mesh, where the upper optical branch has a stable static response and a sizeable D4 w_{eff} shift, about $35.2 \rightarrow 53.1 \text{cm}^{-1}$ in the direct fine calculation.

13 Limitations and outlook

The method is deliberately tensor-free, but it still has clear limits.

1. Long-range polar parts of $\Phi^{(3)}$ are not separated analytically. This is the same limitation as the standard `Spectral.py` d_3 route. LO-TO corrections for the interpolated dynamical matrix are future work.
2. The fine mesh does not create new microscopic information. The anharmonic content is still limited by the coarse supercell; nearby fine wavevectors have correlated stochastic noise. The physical gain is the dense two-phonon continuum and the accurately interpolated harmonic dispersion.
3. The ASR constructions here enforce translational sum rules only, as `Apply_ASR` does. Rotational and Born-Huang invariances are not imposed.
4. Designed kernels assume the relevant anharmonic correlations are resolved inside the coarse Wigner-Seitz cell. If the range reaches the boundary, the correct fallback is the plain estimator or a larger coarse cell.
5. Distributed loading, IR/Raman intensity prefactors for the fine mesh, and off-mesh perturbation wavevectors are engineering extensions. The NUDFT field construction already supports arbitrary wavevectors, so spectral maps along paths are a natural next target.
6. The D4 interpolation for underresolved real-material cells remains the main open implementation item. The report gives the four-leg kernel-factorization route and proves its identities, but the runtime `factor4` kernel has not yet been implemented.

A Derivation of the effective interpolation kernel

We derive Eq. (6) in one dimension (the three-dimensional case is the per-dimension product). Let $u(s)$, $s \in \mathbb{Z}$, be the supercell-periodic configuration, $u(s+L) = u(s)$, and let the windowed field at wavevector $\tilde{q}$ (in units where the phase is $e^{-2\pi i \tilde{q}s}$) be

$$\tilde{x}_W(\tilde{q}) = \frac{1}{\sqrt{L}} \sum_s W(s) e^{-2\pi i \tilde{q}s} u(s), \quad (30)$$

with W supported on one period ($W = 1$ on $[0, L)$ for the plain estimator). Consider the three-field product with windows (z, w, v) on the slots and exact conservation $\tilde{q}_1 + \tilde{q}_2 = \tilde{q}_p$:

$$P = \overline{\tilde{z}(\tilde{q}_p)} \tilde{x}_w(\tilde{q}_1) \tilde{y}_v(\tilde{q}_2) = \frac{1}{L^{3/2}} \sum_{s_1 s_2 s_3} z(s_3) w(s_1) v(s_2) e^{2\pi i (\tilde{q}_p s_3 - \tilde{q}_1 s_1 - \tilde{q}_2 s_2)} \bar{u}(s_3) u(s_1) \delta f(s_2). \quad (31)$$

Taking the ensemble average and using translation invariance, $\langle \bar{u}(s_3) u(s_1) \delta f(s_2) \rangle = c_3(s_1 - s_3, s_2 - s_3)$ with c_3 evaluated at the *actual* integer differences (the physical correlation of the periodic system, which itself is the image sum $c_3(\delta_1, \delta_2) = \sum_{n_1 n_2} c_3^\infty(\delta_1 + n_1 L, \delta_2 + n_2 L)$ of the bulk correlation). Substituting $\delta_i = s_i - s_3$ and using $\tilde{q}_p - \tilde{q}_1 - \tilde{q}_2 = 0$ *exactly*, the s_3 dependence of the phase cancels:

$$\langle P \rangle = \frac{1}{L^{3/2}} \sum_{\delta_1 \delta_2} \underbrace{\left[\sum_{s_3} z(s_3) w(s_3 + \delta_1) v(s_3 + \delta_2) \right]}_{S_{z,w,v}(\delta_1, \delta_2)} c_3(\delta_1, \delta_2) e^{-2\pi i (\tilde{q}_1 \delta_1 + \tilde{q}_2 \delta_2)}. \quad (32)$$

Because the windows are zero-padded (support $\leq L$), the differences δ_i range over $(-L, L)$ and each *physical* image of c_3^∞ is weighted by S at its actual (unwrapped) difference and carries its actual phase — this is where the centering choice lives. Two consequences:

- **Commensurate limit.** If $\tilde{q}_i$ are commensurate, all images of a difference class $(\delta_1, \delta_2) \bmod L$ share the same phase, so only the class sums $\sum_{\mathbf{n}} S(\delta + L\mathbf{n})$ matter. If they equal L (partition of unity), Eq. (32) collapses to the exact coarse average — for *any* window design satisfying the constraint. The plain window satisfies it identically ($S = L$ — spread, class sums = L).
- **Off-grid.** The kernel S/L is the image-assignment rule of the interpolation. The minimal-image target of Sec. 4.2 makes it the indicator of the most compact image (ties split), reproducing zero-padded tensor centering.

The same algebra applies verbatim to every D3 term (the y field can sit in any slot) and, with one more field, to the D4 terms, whose kernel is the window quadruple correlation. Note also that no origin appears in Eq. (32): translating the window (equivalently, rolling the data) leaves the expectation invariant, so origin averaging only reduces variance.

B Gaussian integration by parts and the pre-filter identity

For the Gaussian density ρ_Υ with covariance $\langle u_a u_b \rangle = \Upsilon_{ab}^{-1}$, Wick/Isserlis integration by parts gives, for any observable $F(u)$, $\langle u_a F \rangle = \Upsilon_{ab}^{-1} \langle \partial F / \partial u_b \rangle$. Applying it twice to the D3 average (with $\delta f = f - f_{\text{SSCHA}} - \langle f \rangle$, whose second derivative equals that of the full force $f = -\partial V / \partial R$):

$$\langle u_a u_b \delta f_c \rangle = \underbrace{\Upsilon_{ab}^{-1} \langle \delta f_c \rangle}_{=0} + \Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \left\langle \frac{\partial^2 \delta f_c}{\partial u_{a'} \partial u_{b'}} \right\rangle = -\Upsilon_{aa'}^{-1} \Upsilon_{bb'}^{-1} \langle \partial_{a'b'c}^3 V \rangle. \quad (33)$$

In the SSCHA mode basis Υ^{-1} is diagonal with eigenvalues $f_\psi(\mathbf{q}, \nu) = (1 + 2n_{\mathbf{q}\nu}) / 2\omega_{\mathbf{q}\nu}$: every *displacement* leg of the correlation carries a propagator factor f_ψ , while force legs carry none.

Since the Lanczos kernel multiplies every displacement leg by $f_{\Upsilon} = 1/f_{\psi}$ (the $r_1 = f_{\Upsilon}x$ fields), the combination it averages is the *bare* vertex contraction. The pre-filter of Sec. 6 simply moves the f_{Υ} multiplication *before* the off-grid transform: on the coarse grid $f_{\Upsilon}x$ is computed exactly (mode-diagonal), and its real-space transform is a configuration whose correlations with y are contractions of the short-ranged $\Phi^{(3)}$ (in the Cartesian basis, before any polarization projection — the projections are applied at the fine wavevectors analytically). Off-grid, interpolating the filtered fields therefore interpolates a Φ -ranged object; the exact fine-side factors $f_{\psi}(\tilde{\mathbf{q}})$ that the kernel formulas still require are supplied analytically (folded into the $\alpha^{(1)}$ blocks and set to unity in the kernel tables).

For the four-field averages the same expansion produces, besides the connected $\langle \partial^4 V \rangle$ term, disconnected pieces proportional to $\langle u \otimes \delta f \rangle = \Upsilon^{-1}(\langle \partial^2 V \rangle - \Phi)$, which vanish when Φ is the converged SSCHA matrix. This is the stationarity condition under which the uniform vertex rescaling of Sec. 5 is exact.

C Reproducibility

All numbers and figures in this report are produced by the scripts in `tests/test_interpolation/` (toy model and physics tests) and the benchmark/figure scripts referenced in the repository (`report/interpolation/scripts/`, including `asr_data.py` and `asr_figs.py` for Sec. 7); the decisive physics comparisons are enforced as regression tests (`test_physics_interp.py`, `test_windows.py`, `test_asr_windows.py`). The atom-resolved geometry checks are in `test_atomic_windows.py`. The SnTe oracle and response numbers of Sec. 8 are tracked in `SnTe_FF/Spectral/TDSCHA_Interpolate/issues`; the relevant scripts/data are `kernel_study.py`, `atomic_d3_N300_s20.dat` and `atomic_d3_N1000_s30.dat`.

References

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- [2] TDSCHA q -space Lanczos implementation, `tdscha.QSpaceLanczos` (this repository).
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